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Medellín, Colombia
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ALEXEY VELASQUEZ

AI Engineer | ML Engineer

Self-hosted AI systems — speech intelligence, LLM agents
and interpretable ML, built end to end and run in production.

alexey0424.github.io
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TECHNICAL SKILLS

Languages	Python, SQL, TypeScript.
ML / AI	PyTorch, scikit-learn, LightGBM, XGBoost, PyCaret, TensorFlow, Transformers (BERT), Whisper ASR, speaker diarization, speech emotion recognition, LLMs (Ollama), RAG, embeddings · pgvector, SHAP.
Platforms	PostgreSQL, Docker, FastAPI, n8n, Make.com, Cloudflare Workers, Azure, Databricks, MLflow, Power BI, Git, GitHub Actions (CI).
Data	NumPy, pandas, SciPy, Matplotlib, Seaborn.
Methods	Statistical testing, A/B testing, schema-constrained LLM decoding, MCP (Model Context Protocol), event-driven architecture (webhooks).

EXPERIENCE

Data Scientist

Luis Buys Homes LLC

Jan 2026 – Present

Florida, USA · Remote

- Built a production CRM automation suite wiring Follow Up Boss, JustCall, Slack and Google Forms into **3 event-driven pipelines**; a webhook-first redesign cut **~43,000** idle polling operations/month to **~3** per event, with every lead logged, deduplicated (E.164) and routed to the right closer with zero manual touch.
- Shipped a remote **MCP server** (TypeScript, Cloudflare Workers, Google OAuth) that lets the sales team query the CRM in natural language through Claude — 12 role-gated tools, per-user data scoping enforced server-side, 100% read-only by construction.
- Enriched every logged call with **AI call reports** (summary, call score, topics, customer sentiment) from JustCall webhooks; a single-step write design eliminated a production race condition.
- Containerized **BERT sentiment analysis** over first-contact sales calls (Docker) and delivered PostgreSQL → Power BI dashboards the sales team ran their day on.

Data Scientist

Cactus Home Offer · cactushomeoffer.com

Jun 2025 – Dec 2025

Florida, USA · Remote

- Built an end-to-end **speech intelligence** pipeline — Whisper ASR, speaker diarization, speech-emotion recognition and SHAP interpretability — generating per-segment behavioral insights from real-estate sales calls: up to **80% less manual call review** and measurably improved closer performance.
- Replaced rule-based KPI extraction with **schema-constrained LLM extraction** (objections, buying signals, rapport quality, call outcome) over mixed Spanish/English transcripts, eliminating keyword logic.
- Integrated CRM-triggered **real-time inference** through n8n, leveraging Azure-hosted services and Databricks API integrations to automate sales coaching.

Machine Learning Engineer

EIA University

Aug 2023 – Aug 2025

Medellín, Colombia

- Developed vibration-based structural damage detection for a real 14 m wind turbine: LightGBM and XGBoost over modal features from **6,000+** field records, reaching up to **99.5% detection accuracy** with SHAP explanations validated against the structure's physics.
- Ran comparative model evaluation (A/B testing) across Random Forest, SVM, LightGBM and autoencoders, and designed PostgreSQL ingestion for continuous real-time monitoring across a 1.5-year experimental campaign.

OPEN-SOURCE AI SYSTEMS — ALEXEY0424.GITHUB.IO

- ORCA call intelligence** — self-hosted GPU stack (Whisper, pyannote, emotion2vec, Ollama, pgvector RAG): raw sales call in, coaching scorecard and chat-searchable archive out; nothing leaves the machine.
- LLM web scraper** — plain English in, verified spreadsheet out; every extracted value grounded against the source page (151 offline tests). **+5 more public systems** on GitHub.

EDUCATION

Master's in Engineering, Machine Learning focus — 1st Class (94%), EIA University.

Aug 2025